



MedX

User Manual

The complete guide to running your pathology lab with MedX — from patient registration and GST billing to verified reports, WhatsApp delivery, and the online patient portal.

Document	Audience	Version	Date
User Manual	Lab owners, receptionists, technicians, pathologists	1.0 (App v0.1.0)	15 July 2026

MedX — Offline-first Pathology Lab Management for Indian Labs

Contents

- 1. Welcome to MedX**
- 2. Installation & First Launch**
- 3. Your License & Plans**
- 4. Users, Roles & Permissions**
- 5. Front Desk: Dashboard, Queue & Tokens**
- 6. Registering Patients & Creating Orders (Billing)**
- 7. Sample Tracking, Barcodes & Receipts**
- 8. Result Entry & Pathologist Verification**
- 9. Reports: Print, PDF & WhatsApp Delivery**
- 10. Payments, Invoices & Outstanding Dues**
- 11. Appointments & Home Collection (incl. Online Bookings)**
- 12. Doctors, Commissions & Corporate Billing (Pro)**
- 13. Inventory: Reagents & Suppliers (Pro)**
- 14. Machine Interfacing — Auto-import from Analyzers (Pro)**
- 15. Quality & NABL Compliance Modules (Enterprise)**
- 16. The Patient Website & Cloud Sync**
- 17. Counter Display for Waiting Areas**
- 18. Backup & Data Safety**
- 19. Troubleshooting & FAQ**

1. Welcome to MedX

MedX is a complete laboratory management system built specifically for Indian pathology labs. It runs as a Windows desktop application that works **fully offline** — every patient record, bill, and report is stored locally on your lab's PC, so day-to-day work never depends on the internet. Higher plans add online superpowers: a patient report portal, online home-collection bookings, analyzer machine connections, and more.

What you can do with MedX

- Register patients and create GST-compliant bills in under a minute.
- Track samples with barcoded accession numbers from collection to report.
- Enter results with automatic High/Low/Critical flags based on age- and sex-specific Indian reference ranges.
- Lock reports behind pathologist verification, then print a professional A4 report or save a PDF.
- Send reports to patients on WhatsApp with one click (Pro plan).
- Give patients a self-service website to download reports and book home collections (Pro plan).
- Connect lab analyzers so results flow in automatically (Pro plan).
- Run NABL-style quality logs, temperature charts, calibrations and SOP registers (Enterprise plan).

Note: MedX stores money values exactly (in paise) — totals, GST and dues never suffer rounding errors.

2. Installation & First Launch

Installing

- Run **MedX-Setup-0.1.0.exe** (your vendor sends this file). The wizard installs MedX and places a shortcut on the Desktop and Start Menu.
- A portable version (**MedX — Pathology LIMS 0.1.0.exe**) is also available — it runs without installation, e.g. from a USB drive.
- Reinstalling a newer version over an old one keeps all your data and your license — nothing needs to be re-entered.

First-time setup

- Open **Settings** → **Lab Profile** and fill in your lab name, address, phone, email, GSTIN and state code. This appears on every invoice and report.
- Add your staff in **Settings** → **Users** and give each person the right role (see Section 4).
- If you purchased a plan, paste your license key in **Settings** → **System License** (see Section 3).

Important: The lab name is locked once a license is active — it comes from your license and cannot be edited. This prevents misuse of the software under a different lab's name. Contact your vendor if your lab name changes.

3. Your License & Plans

MedX is sold in three plans. Every feature in the app automatically unlocks or locks based on the license you activate — there is nothing else to configure.

Plan	Best for	Devices	Highlights
Starter	Solo labs starting out	1 PC	Complete daily workflow: registration, GST billing, results, verified A4 reports, barcode labels, thermal receipts, token queue, backups. Fully offline; 15 patients/day cap.
Pro	Growing labs	3 PCs	Everything in Starter, unlimited patients, plus: WhatsApp report delivery, online patient portal & home-collection bookings, analyzer machine interfacing, doctor commissions, inventory, analytics.
Enterprise	Multi-branch / NABL labs	Unlimited	Everything in Pro plus NABL quality modules (QC logs, calibrations, temperature, SOPs, audit trail), TPA claims, ABDM/ABHA integration, MIS reports.

Activating your license

- Your vendor sends you a long text token. Open **Settings** → **System License**, paste it, and press **Activate License**.
- Activation works offline — the app verifies the token's cryptographic signature locally. When internet is available, the app also checks in with the vendor.
- The license card shows your plan, lab name, registered phone and expiry date.

Staying up to date automatically

While online, MedX quietly checks in with your vendor every 5 minutes and at startup. Through this check-in, the following happen **automatically** — you never re-enter a key:

- **Renewals:** when you pay for another period, the new expiry date appears on its own.
- **Plan upgrades:** upgrading Starter → Pro instantly unlocks the new modules.
- **Vendor messages:** notices (e.g. payment reminders, maintenance alerts) appear as a yellow banner at the top of the app. Press **X** to dismiss.
- You can force an immediate check with **Settings** → **System License** → “**Check License Status Now**”.

Device limits

Each plan allows a fixed number of PCs. A PC registers itself the first time it checks in. If you replace a computer, ask your vendor to free the old device's slot — it takes them one click.

Important: If the license is revoked by the vendor (e.g. non-payment), the app reverts to the free Starter tier at the next check-in. Your data is never touched — only paid features lock.

4. Users, Roles & Permissions

MedX ships with four roles, each seeing only the pages they need. The active user is switched from the dropdown at the top-right of the app.

Role	Default access
Owner	Everything, including Settings, Users and license management.
Receptionist	Front desk, patient registration, billing, appointments, report list & delivery.
Technician	Sample tracking, result entry, inventory, QC and machine interfacing.

Role	Default access
Pathologist	Result entry, verification, QC logs and SOPs.

- Owners can customise which pages every role sees: **Users** → **Role Permissions** tab. Tick or untick pages per role; changes save instantly.
- Switching to the Owner account requires the administrator passcode, so staff cannot self-upgrade their access.
- Only pages allowed by your license tier appear in the permission grid.

5. Front Desk: Dashboard, Queue & Tokens

Dashboard

The dashboard shows today's registrations, collected payments, pending dues and the live worklist — a one-glance health check of the day.

Queue / Tokens

- Press **Issue Token** to hand a waiting patient a token like **T-007** (patient name optional, purpose selectable).
- Press **■ Call Next** to serve the next waiting token; the current one is marked done.
- **Skip** removes a token from the queue (e.g. patient left).
- Token numbers restart from T-001 every day automatically.

6. Registering Patients & Creating Orders (Billing)

New Patient

- Fill name, age/DOB, sex and phone. The phone number powers returning-patient lookup and portal access, so enter it carefully.
- For a returning patient, just type their phone number — their record appears instantly.

New Order (the bill)

- Search any of ~2,500 tests and profiles by name or code, or use one-tap health packages.
- Prices come from your editable catalog; apply per-item or whole-bill discounts as needed.
- GST is computed automatically (diagnostic services are exempt by default under SAC 999316; taxable items compute CGST/SGST or IGST from your state code).
- Choose the payment mode — Cash, UPI, Card, or Due (partial payments supported).
- Saving the order generates the invoice number (financial-year series, e.g. INV/26-27/0042) and an accession number for the sample.
- Print the A4 invoice or the 58 mm thermal receipt immediately from the order screen.

Estimates

Quote prices without registering a patient: build a test list in **Estimates**, show the total, and convert it into a real order when the patient agrees.

7. Sample Tracking, Barcodes & Receipts

- Every order carries an accession number like **260715-0001** (date + serial).
- Print **barcode label sheets** from the order view — stick them on tubes; scanners and analyzers read them directly.
- Track each sample's stage in **Sample Tracking**: Pending → Collected → Received → Processed.
- The 58 mm thermal receipt is designed for standard POS printers.

8. Result Entry & Pathologist Verification

- Open a pending order in **Result Entry**; every analyte shows its unit and the correct reference range for the patient's age and sex.
- As you type values, **H** (high), **L** (low) and **critical** flags appear live.
- Results entered by machine interfacing (Section 14) appear automatically with an audit log entry.
- A report can only be released after a **Pathologist verifies** it in the Verification page — verification locks the results and records who verified and when.

9. Reports: Print, PDF & WhatsApp Delivery

- The A4 report includes your letterhead, GSTIN, reference ranges, H/L flags and the pathologist's signature block.
- ■ **Download PDF** creates a polished PDF — available on the order's Report tab, the Report List, and Delivery Status pages.
- **WhatsApp (Pro)**: one click opens WhatsApp with the patient's number and a ready message — sent from *your lab's own* WhatsApp, so there are no per-message charges.
- **Delivery Status** tracks which reports have been delivered and via which channel (WhatsApp / print / email).

10. Payments, Invoices & Outstanding Dues

- **Invoices** lists every bill with totals, GST breakup and payment state.
- **Payments** records every receipt with mode and reference number.
- **Outstanding** shows all dues at a glance — collect a due and the balance updates everywhere instantly.

11. Appointments & Home Collection

Appointments

The calendar view schedules visits with date, time, patient and purpose — statuses move from Scheduled to Arrived to Done.

Home collection visits

- Schedule phlebotomist visits in **Home Collection** with date, slot, address and assigned phlebotomist.
- Progress each visit: Scheduled → Collected → Received (sample reached the lab) or Cancelled.

Online bookings from your website (Pro)

- Patients can book home collections on the MedX patient website. Bookings arrive in your app **automatically within 5 minutes**, or instantly when you press ■ **Fetch Online Bookings**.
- Imported bookings appear under Upcoming with the note “Booked online via MedX website”, plus the patient's age and requested tests.
- Once imported, the booking is removed from the cloud — your PC is its only home.

12. Doctors, Commissions & Corporate Billing (Pro)

- **Doctor List:** referring doctors/clinics with their commission configuration.
- **Commission:** referral earnings computed per doctor from actual billed orders; record payouts when you settle.
- **MOU Clients & Corporate Billing:** manage tie-ups with special rate lists and monthly consolidated bills.

13. Inventory: Reagents & Suppliers (Pro)

- Track reagents and consumables with units, stock levels and expiry dates.
- Record stock movements in and out; low-stock items stand out so you reorder in time.
- Keep a supplier directory with contact details for quick reordering.

14. Machine Interfacing — Auto-import from Analyzers (Pro)

MedX listens on TCP port **8100** for data from lab analyzers (hematology counters, biochemistry analyzers, etc.) speaking the industry-standard **ASTM** or **HL7** protocols.

- Configure your analyzer (or its middleware PC) to send results to the MedX PC's IP address, port 8100.
- When a frame arrives, MedX matches the sample barcode to the order and fills the result values automatically, marks the sample processed, and logs the event.
- The **Machine Interfacing** page shows a live connection indicator, a log of received frames, and a built-in **simulator** so you can test the whole flow without a physical machine.
- Serial-port (RS-232) analyzers are supported via the serial connection panel on the same page.

15. Quality & NABL Compliance Modules (Enterprise)

- **QC Logs:** daily control runs with Westgard rule evaluation.
- **Temperature:** fridge/freezer temperature register with out-of-range highlighting.
- **Calibrations:** instrument calibration schedule and history.
- **SOPs:** a versioned register of standard operating procedure documents.
- **Audit:** an automatic trail of every important action — who did what, and when.

16. The Patient Website & Cloud Sync

Labs on the Pro or Enterprise plan get a presence on the MedX patient website (<https://medx-web-one.vercel.app>):

- **Report portal:** patients enter their invoice number + registered phone to view and download their full report history. Both must match — nobody else can see their reports.
- **Home collection booking:** patients pick your lab, fill their details, and the booking flows into your app (Section 11).
- **Tests & pricing:** your catalog with prices is browsable so patients know costs upfront.

How sync works

- While online, your app securely pushes orders and patient records to the portal every 5 minutes and shortly after any change.
- The sidebar's footer shows the sync state: **green “Cloud Data Synced”** or red “Pending Cloud Sync”.
- Every push is authenticated with your license's cryptographic token — nobody can impersonate your lab.

Note: Starter-plan labs are 100% offline by design: no patient data ever leaves the lab PC, and the portal does not list the lab. This is a privacy guarantee, not a limitation.

17. Counter Display for Waiting Areas

- From **Queue / Tokens**, press **■ Open Counter Display** — a clean full-screen “NOW SERVING T-00X” board opens in a second window.
- Drag it onto a TV or second monitor facing the waiting area and press F11 for fullscreen if desired.
- It updates **live**: the moment reception presses Call Next, the board flips to the new token.
- The display window is read-only — it cannot change any data.

18. Backup & Data Safety

- All data lives locally in the MedX database on your PC — it works with no internet, forever.
- **Settings** → **Export data (JSON)** downloads a complete backup file. Keep a copy on a USB drive or another PC weekly.
- The Backup page lets you configure scheduled automatic backups to a folder of your choice.
- Reinstalling or updating MedX never deletes your data.

19. Troubleshooting & FAQ

Problem	Solution
The app opens with a white screen	Reinstall using the latest MedX-Setup exe from your vendor — this was fixed in a newer build.
License won't activate	Make sure you pasted the entire token (it is long). Tokens from before a security update are invalid — ask your vendor for a fresh key.
“Device limit exceeded” on activation	Your plan's PC slots are full. Ask your vendor to remove a retired device or upgrade the plan.

Problem	Solution
Renewed, but expiry date looks old	Open Settings → System License → “Check License Status Now” while online; the new date applies immediately.
Online booking not showing in the app	Home Collection → “  Fetch Online Bookings”. Requires an active Pro/Enterprise license and internet.
Patient can't find their report on the portal	Verify the invoice number and that the phone matches the one registered on the bill. The lab PC must have synced (green dot) after the report was verified.
Counter display frozen	Update to the latest build — displays now update live. Close and reopen the display window after updating.
Machine data not importing	Confirm the analyzer targets the MedX PC's IP on port 8100 and the barcode on the sample matches the accession number.
Wrong staff seeing too many pages	Owner → Users → Role Permissions: untick pages per role.
I need my data on a new PC	Export data (JSON) on the old PC, install MedX on the new one, and ask your vendor to move the device slot.

Need help? Contact your MedX vendor — they can see your license status, connected devices and last sync time in real time, and can send fixes directly to your app.